Predictive Analytics for Workforce Stability: Identifying Attrition Factors

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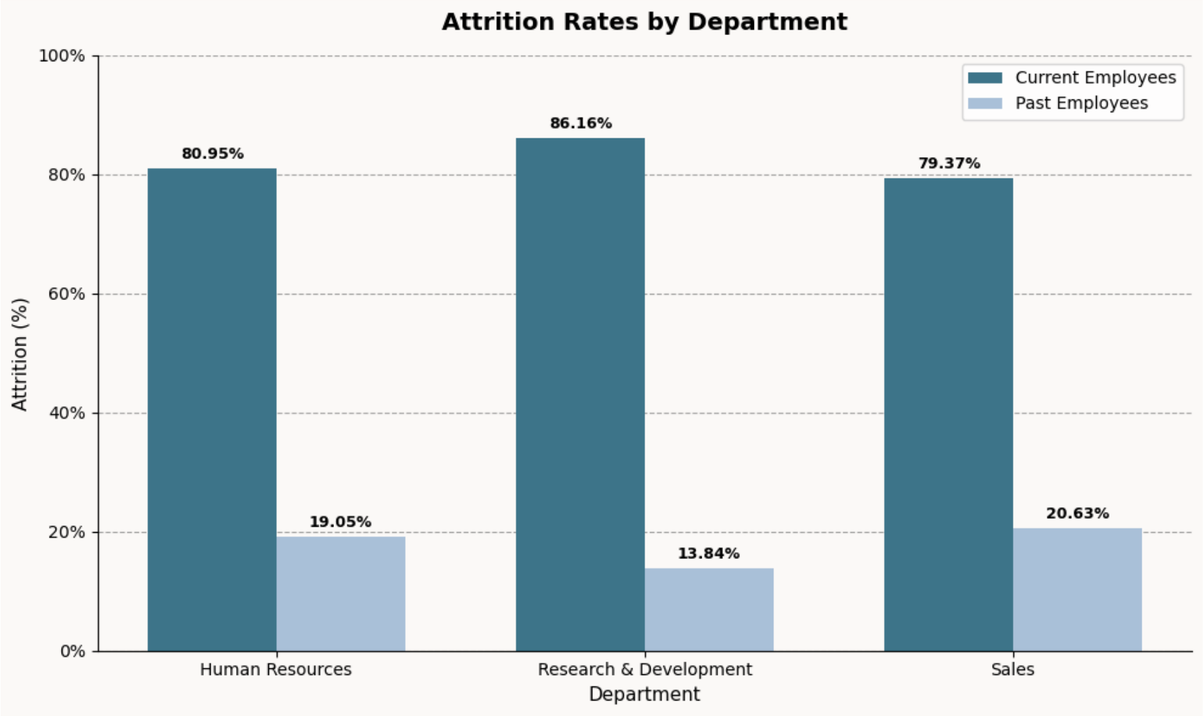
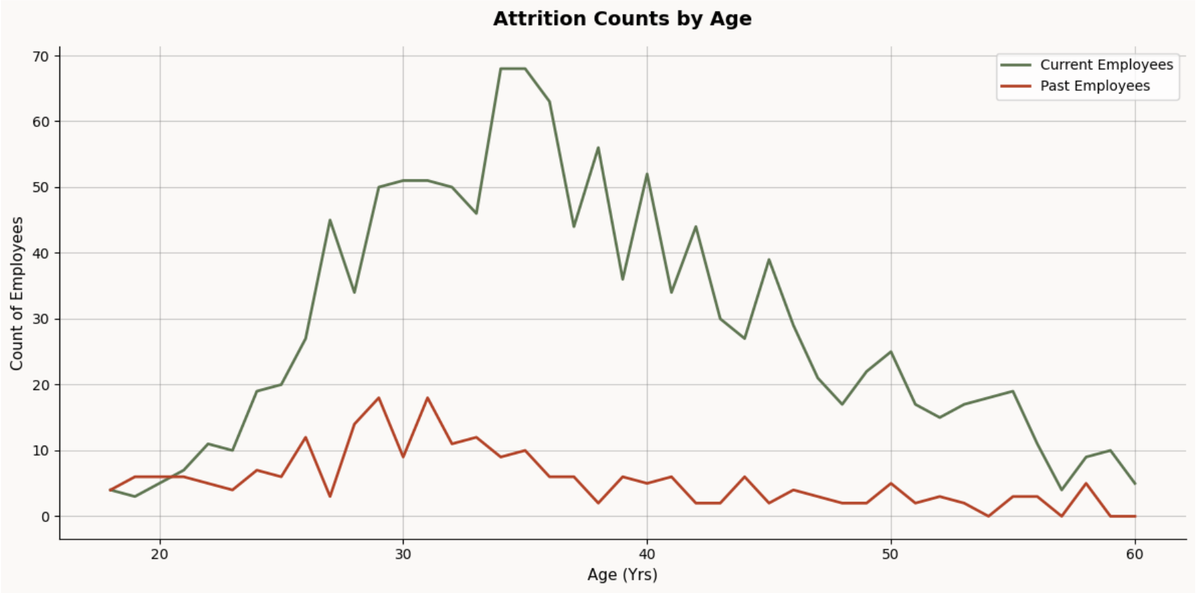
What is employee attrition?

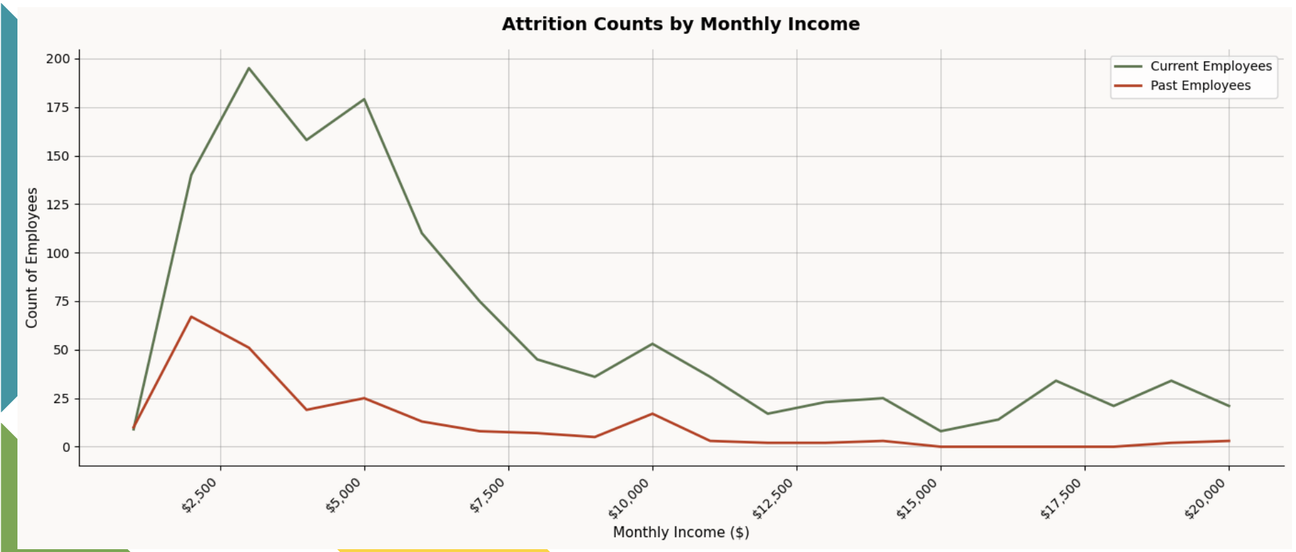
- Gradual lowering of workers within a company and their positions are not immediately replaced
- Types:
  - Voluntary leave
    - Quitting
    - Retirement
  - Involuntary leave
    - Layoffs
    - Termination
- High employee attrition leads to increased hiring cost & reduced team productivity

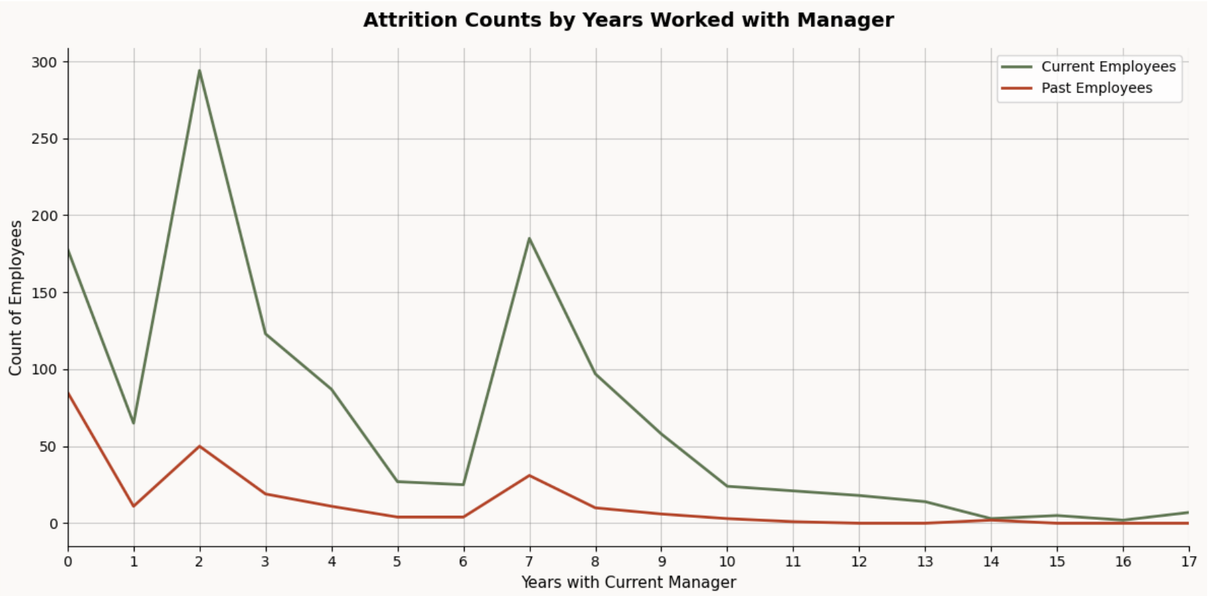
Employee Data & Ethical Concerns

- Dataset: synthetic HR Employee Attrition & Performance dataset from IBM
- Real employee data is difficult to obtain, companies do not make personnel records public and accessing non-synthetic demographic data typically requires an NDA making it inaccessible for open research
- Anonymized personal data is vulnerable to re-identification
  - Netflix public dataset challenge: researchers were able to reverse-engineer anonymized customer records to identify individuals

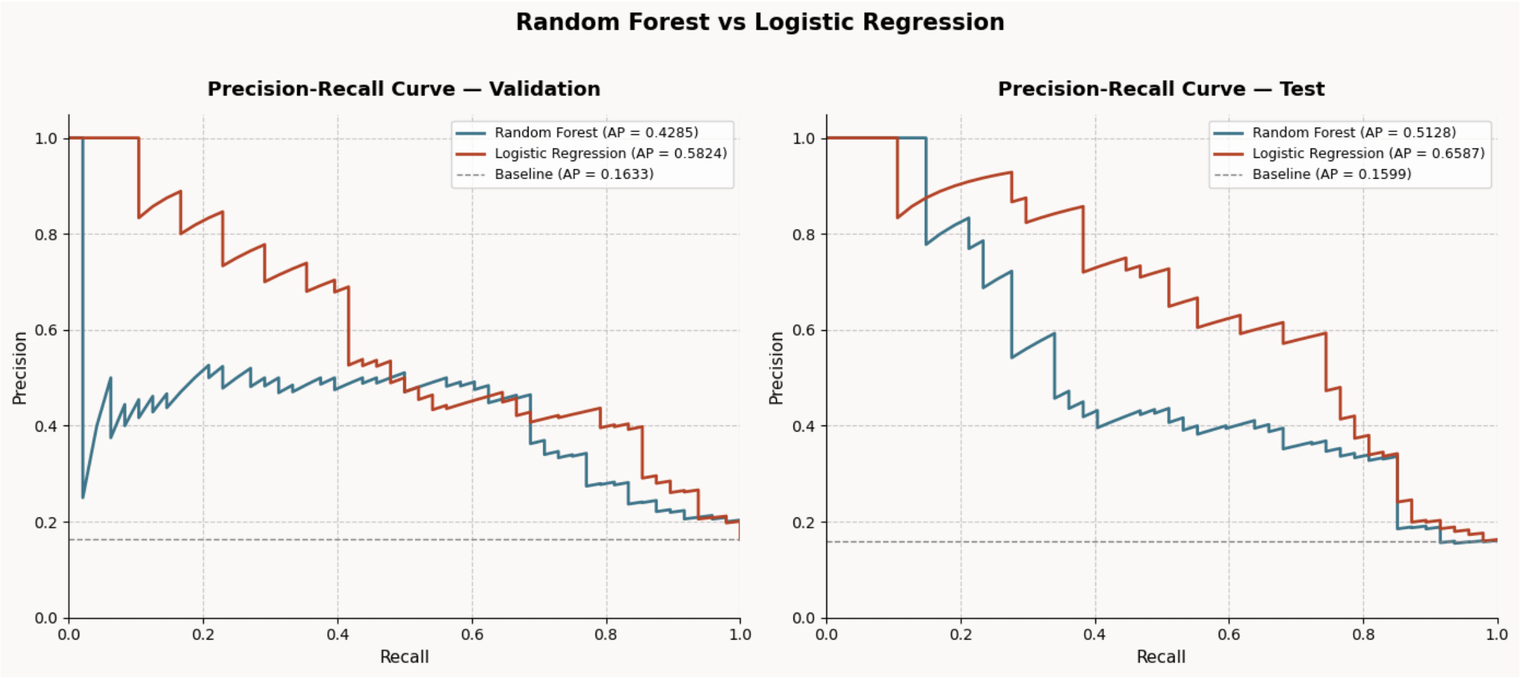

Attrition Rates by Department
Current Employees
Past Employees
100%
80%
60%
40%
20%
0%
Attrition (%)
80.95%
19.05%
86.16%
13.84%
79.37%
20.63%
Human Resources
Research & Development
Sales
Department

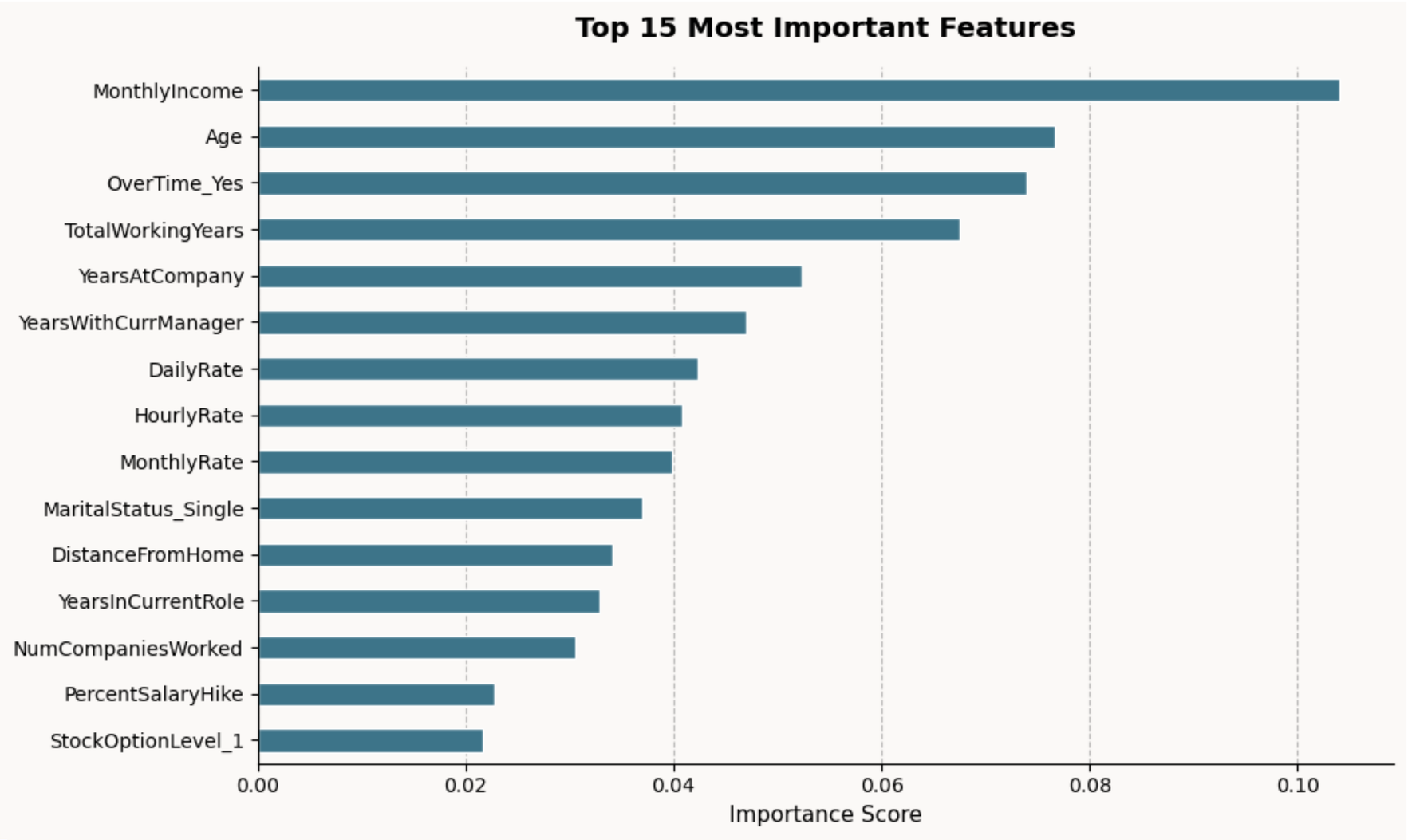

Attrition Counts by Age
Current Employees
Past Employees
Count of Employees
Age (Yrs)


Attrition Counts by Monthly Income
Current Employees
Past Employees
Count of Employees
$2,500
$5,000
$7,500
$10,000
$12,500
$15,000
$17,500
$20,000
Monthly Income ($)


Attrition Counts by Years Worked with Manager
Current Employees
Past Employees
Count of Employees
Years with Current Manager

Logistic regression vs Random forest




Top 15 Most Important Features
MonthlyIncome
Age
OverTime_Yes
TotalWorkingYears
YearsAtCompany
YearsWithCurrManager
DailyRate
HourlyRate
MonthlyRate
MaritalStatus_Single
DistanceFromHome
YearsInCurrentRole
NumCompaniesWorked
PercentSalaryHike
StockOptionLevel_1
0.00
0.02
0.04
0.06
0.08
0.10
Importance Score

Findings

- Money matters!!
- Younger people (18-20) tend to leave the company more than any other ages/age group.
- Employees in departments with more KPI focus like Sales tend to leave more over employees who work in administrative settings like Human Resources.

References

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